ITA0610-Machine Learning

LAB EXPERIMENTS

Vasantha Nithi D Y

192424137

Experiment 7: Logistic Regression

Aim

To implement the Logistic Regression algorithm in Python for binary classification and evaluate its accuracy.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the Breast Cancer dataset.

Step 3: Split the dataset into training and testing sets.

Step 4: Create a Logistic Regression model.

Step 5: Train the model using the training data.

Step 6: Predict the class labels for the test data.

Step 7: Calculate and display the accuracy of the model.

Program

```
#Experiment 7: Logistic Regression
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

data = load_breast_cancer()

X_train, X_test, y_train, y_test = train_test_split(
    data.data,
    data.target,
    test_size=0.3,
    random_state=0
)

model = LogisticRegression(max_iter=5000)

model.fit(X_train, y_train)

prediction = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, prediction))
```

Output

```
•••  Accuracy: 0.9590643274853801
```

Result

Thus, the Logistic Regression algorithm was successfully implemented, and the model classified the test data with high accuracy.